ICSE

ANALYSIS OF PUPIL PERFORMANCE

October 2025

MATHEMATICS

Research Development and Curriculum Division
Council for the Indian School Certificate Examinations
New Delhi

OCTOBER 2025

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Council for the Indian School Certificate Examinations (CISCE)

MISSION STATEMENT

The Council for the Indian School Certificate Examinations is committed to serving the nation's children, through high quality educational endeavours, empowering them to contribute towards a humane, just and pluralistic society, promoting introspective living, by creating exciting learning opportunities, with a commitment to excellence.

ETHOS OF CISCE

- Trust and fair play.
- Minimum monitoring.
- Allowing schools to evolve their own niche.
- Catering to the needs of the children.
- Giving freedom to experiment with new ideas and practices.
- Diversity and plurality - the basic strength for evolution of ideas.
- Schools to motivate pupils towards the cultivation of:
  Excellence - The Indian and Global experience.
  Values - Spiritual and cultural - to be the bedrock of the educational experience.
- Schools to have an 'Indian Ethos', strong roots in the national psyche and be sensitive to national aspirations.

FOREWORD

The Council for the Indian School Certificate Examination has endeavoured to sincerely take forward the vision of NEP 2020 in enhancing pedagogy and assessments through systemic reforms. In alignment with this vision, the CISCE also remains committed to ensuring that learning is meaningful, and that assessment practices are utilised as tools for growth, reflection, and continuous improvement rather than as an exercise in mere evaluation.

The ISC and ICSE examinations have long served as the cornerstone of the CISCE's academic framework, especially in shaping the educational journeys and futures of countless CISCE students. In this context, the *Analysis of Pupils' Performance* documents have played a significant role in strengthening the CISCE's assessment culture. These reports have, over the years, supported students in understanding subject-wise expectations and in preparing effectively for the Board examinations.

Our heartfelt appreciation is extended to the Research Development and Curriculum Division (RDCD) of the CISCE for their dedicated efforts in preparing this detailed document for the benefit of students. Our sincere appreciation and gratitude to the subject-matter experts at School of Continuing Education and University Resource Centre, Azim Premji University (APU) for their comprehensive and thoughtful analysis of select ICSE subject documents. We also express our gratitude to the examiners for their meticulous evaluation and insightful feedback on candidates' performance, as well as for recommending pedagogical strategies to help teachers address common areas of difficulty.

We are confident that students, teachers, and parents will make the best use of this document by engaging with its various sections in depth and by implementing the learnings to ensure enhanced and successful performance in the forthcoming examinations.

October 2025

Dr. Joseph Emmanuel
Chief Executive & Secretary
CISCE

PREFACE

The *Analysis of Pupil Performance* document is a standing legacy of the Council for Indian School Certificate Examinations (CISCE) that aims at capturing performance trends of CISCE candidates in the annually conducted ICSE and ISC examinations. Over the years, these user-friendly and accessible reports have helped CISCE schools analyse patterns in candidates' responses to board examination papers, while also providing valuable pedagogical insights and suggestions for teachers.

Each year, CISCE undertakes this exercise of a detailed analysis of the performance of candidates for selected subjects at both the ICSE and ISC examinations. Each subject document includes the questions from the board examination along with the marking scheme for each question, enabling teachers and students to better understand the scope of the question and the appropriate approach to answering it. It also presents an infographic presentation of statistical analysis and contextualised commentaries on candidate performance with detailed insights and remedies on the errors. Towards the end of the document, a comprehensive summary highlights the topics revealing gaps in conceptual understanding, areas that candidates found confusing, and tailored suggestions to help students approach examinations more effectively.

The uniqueness of this document lies in its micro-analysis of items in board examination papers where suggested pedagogical strategies are tied to each common errors identified from candidates' answers, a strategy that CISCE believes is rudimentary in driving improvements rather than merely sharing a broad stroke macro analysis of candidate performance. While these reports are grounded in real candidate data, they also lend themselves to seamlessly to classroom action.

The Analysis of Pupil Performance document for ICSE for the Examination Year 2025 covers the following subjects - English Language, Literature in English, Hindi, History and Civics, Geography, Mathematics, Physics, Chemistry, Biology, Commercial Studies, Economics, Computer Applications, Economic Applications, Commercial Applications, Environmental Science and Home Science. In addition to the comments by CISCE examiners, the document has also included insights and pedagogical suggestions from experts at School of Continuing Education and University Resource Centre, Azim Premji University (APU).

The subjects covered in the ISC Analysis of Pupil Performance document for the Year 2025 are - Accounts, English Language, Literature in English, Hindi, Economics, Commerce, Business Studies, Mathematics, Physics, Chemistry, Biology, Elective English, History, Political Science, Geography, Psychology, Sociology, Computer Science, Environmental Science, Home Science and Legal Studies.

I extend my heartfelt gratitude to all the ICSE and ISC examiners who have shared their valuable comments on each question. My sincerest appreciation and gratitude to subject matter experts at Azim Premji University (APU) for their detailed analysis of the documents for ICSE subjects. I also acknowledge the efforts of the RDCD team of Dr. Manika Sharma, Ms. Parul Kohli, Ms. Lyimee Saikia, Ms. Mansi Guleria, Ms. Gunjan Khurana, Ms. Madhusree Chatterjee and Ms. Aakriti Agrawal for their commitment to ensuring the compilation of the comments and data into its current form.

We hope that the document is successful in its intended mission of better assisting our teachers in identifying areas that require focused intervention and allowing them to better support CISCE students. At the same time, we also hope that it guides our students in developing a roadmap of effective study methods for their examinations.

October 2025

Dr. Bhawna Taragi
Deputy Head
RDCD, CISCE

QUANTITATIVE ANALYSIS

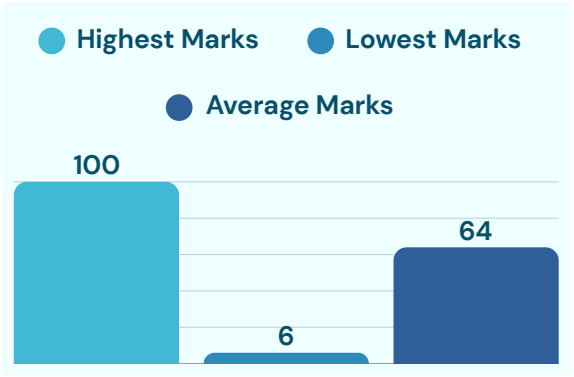
Mathematics

Number of Candidates Appeared
240954

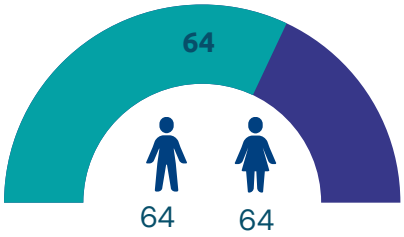
Boys 128976

Girls 111978

Maximum Marks 100



Overall Performance (Mean Marks)

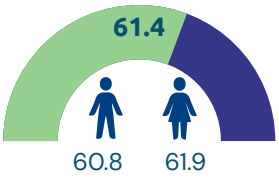


Performance by Gender

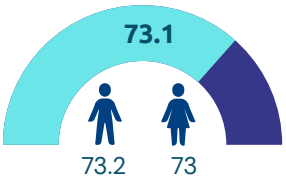
Gender	Mean	SE	t-value
Girls	64	0.06	00*
Boys	64	0.06	

*Not Significant

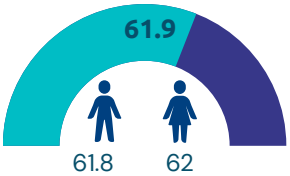
Region-wise Performance (Mean Marks)



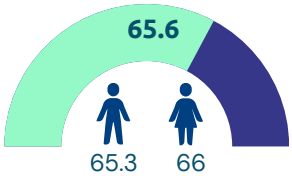
Eastern



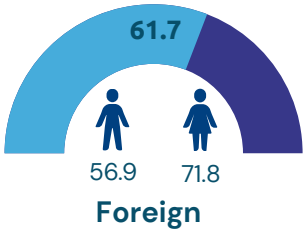
Western



Northern

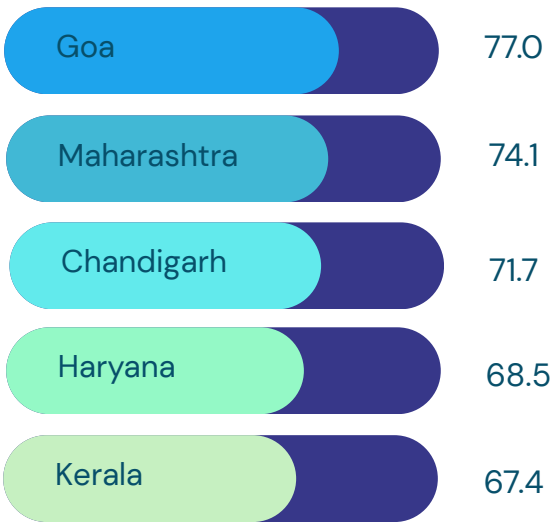


Southern

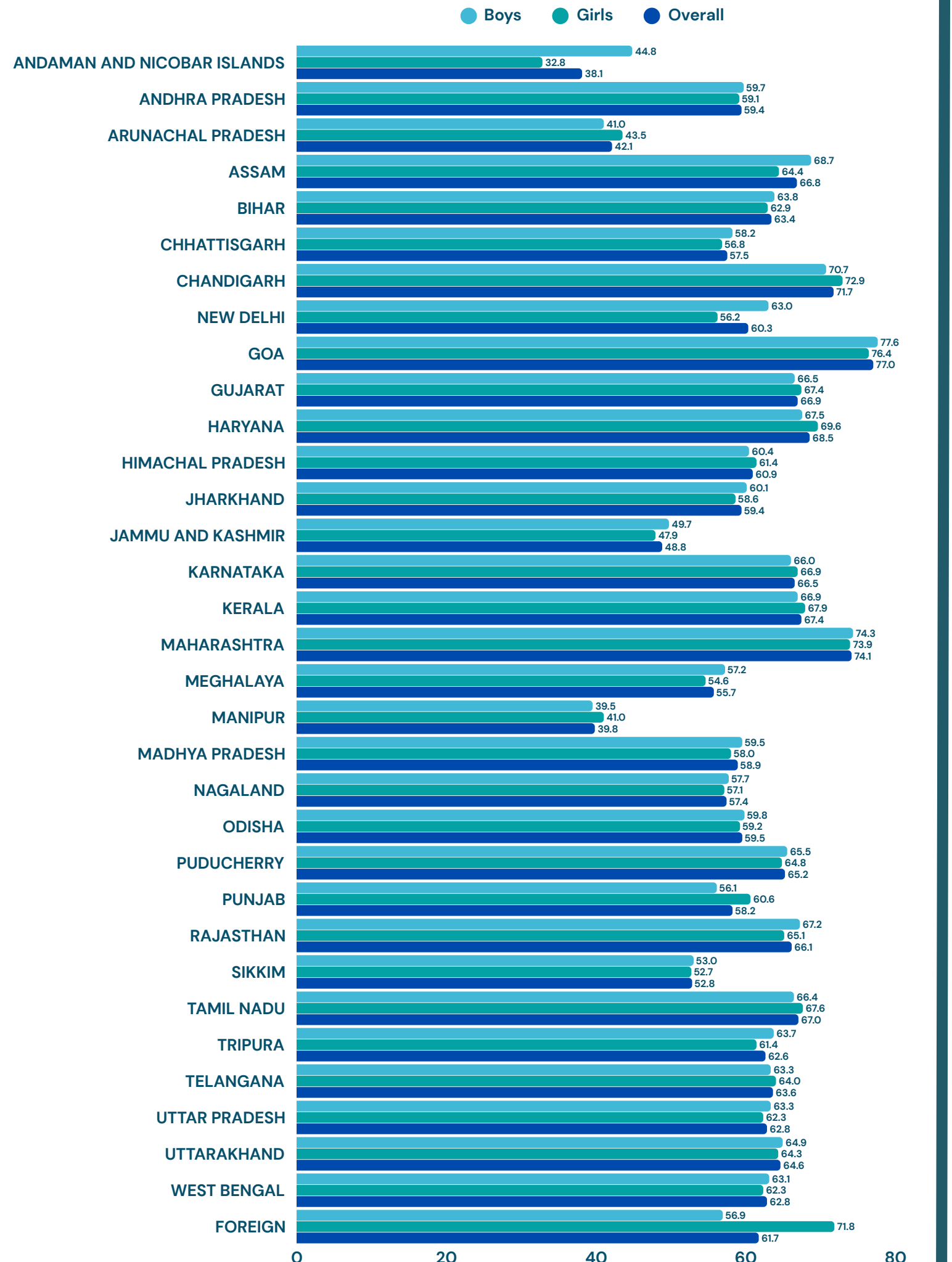


Foreign

Top 5 Perfoming States/UTs (Mean Marks)



Performance State-wise and Foreign (Mean Marks)



SECTION A (40 MARKS)

*(Attempt **all** questions from this **Section**.)*

Question 1

Choose the correct answers to the questions from the given options. **[15]**

(Do not copy the questions, write the correct answers only.

(i) The given quadratic equation $\mathbf{3x^2 + \sqrt{7}x + 2 = 0}$ has:

(a) two equal real roots.

(b) two distinct real roots.

(c) more than two real roots.

(d) no real roots.

(ii) Mr. Anuj deposits **₹500** per month for **18** months in a recurring deposit account at a certain rate. If he earns **₹570** as interest at the time of maturity, then his matured amount is:

(a) ₹(500 × 18 + 570)

(b) ₹(500 × 19 + 570)

(c) ₹(500 × 18 × 19 + 570)

(d) ₹(500 × 9 × 19 + 570)

(iii) Which of the following **cannot** be the probability of any event?

(a) $\frac{5}{4}$

(b) 0.25

(c) $\frac{1}{33}$

(d) 67%

(iv) The equation of the line passing through origin and parallel to the line $\mathbf{3x + 4y + 7 = 0}$ is:

(a) $3x + 4y + 5 = 0$

(b) $4x - 3y - 5 = 0$

(c) $4x - 3y = 0$

(d) $3x + 4y = 0$

(v) If $\boldsymbol{A} = \begin{bmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{bmatrix}$, then $\boldsymbol{A^2}$ is equal to:

(a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

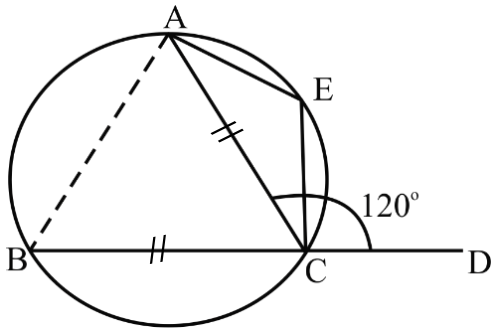
(vi) In the given diagram, chords **AC** and **BC** are equal. If $\angle\mathbf{ACD} = \mathbf{120°}$, then $\angle\mathbf{AEC}$ is:

(a) 30°

(b) 60°

(c) 90°

(d) 120°



(vii) The factor **common** to the two polynomials $\boldsymbol{x^2 - 4}$ and $\boldsymbol{x^3 - x^2 - 4x + 4}$ is:

(a) $(x + 1)$

(b) $(x - 1)$

(c) $(x + 2)$

(d) $(x - 2)$

(viii) A man invested in a company paying **12%** dividend on its share. If the percentage return on his investment is **10%**, then the shares are:

(a) at par

(b) below par

(c) above par

(d) cannot be determined

(ix) **Statement 1:** The point which is equidistant from three non-collinear points **D**, **E** and **F** is the **circumcentre** of the **ΔDEF**.

Statement 2: The **incentre** of a triangle is the point where the bisector of the angles intersects.

(a) Both the statements are true.

(b) Both the statements are false.

(c) Statement 1 is true, and Statement 2 is false.

(d) Statement 1 is false, and Statement 2 is true.

(x) **Assertion(A):** If $\boldsymbol{sin^2A + sinA = 1}$ then $\boldsymbol{cos^4A + cos^2A = 1}$

Reason(R): $\boldsymbol{1 - sin^2A = cos^2A}$

(a) (A) is true, (R) is false.

(b) (A) is false, (R) is true.

(c) Both (A) and (R) are true, and (R) is the correct reason for (A).

(d) Both (A) and (R) are true, and (R) is the incorrect reason for (A).

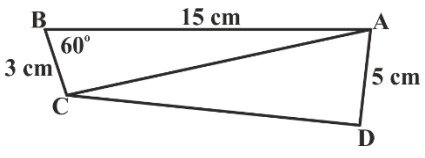
(xi) In the given diagram **ΔABC ~ ΔEFG**. If **∠ABC = ∠ EFG = 60°**, then the length of the side **FG** is:

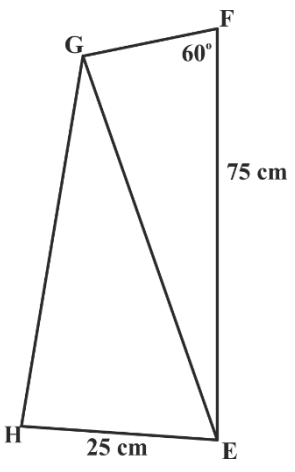
(a) 15 cm

(b) 20 cm

(c) 25 cm

(d) 30 cm





(xii) If the volume of two spheres is in the ratio **27 : 64**, then the ratio of their **radii** is:

(a) 3 : 4

(b) 4 : 3

(c) 9 : 16

(d) 16 : 9

(xiii) The marked price of an article is **₹1375**. If the **CGST** is charged at a rate of **4%**, then the price of the article **including GST** is:

(a) ₹55

(b) ₹110

(c) ₹1430

(d) ₹1485

(xiv) The solution set for $\mathbf{0} < -\frac{\boldsymbol{x}}{\mathbf{3}} < \mathbf{2}, \boldsymbol{x} \in \boldsymbol{Z}$ is:

(a) $\{-5, -4, -3, -2, -1\}$

(b) $\{-6, -5, -4, -3, -2, -1\}$

(c) $\{-5, -4, -3, -2, -1, 0\}$

(d) $\{-6, -5, -4, -3, -2, -1, 0\}$

(xv) **Assertion(A):** The mean of first **9** natural numbers is **4.5**.

Reason(R): Mean = $\frac{Sum\ of\ all\ the\ observations}{Total\ number\ of\ observations}$

(a) (A) is true, (R) is false.

(b) (A) is false, (R) is true.

(c) Both (A) and (R) are true, and (R) is the correct reason for (A).

(d) Both (A) and (R) are true, and (R) is the incorrect reason for (A).

MARKING SCHEME	
(i)	(d) no real roots.
(ii)	(a) ₹(500 × 18 + 570)
(iii)	(a) $\frac{5}{4}$
(iv)	(d) $3x + 4y = 0$
(v)	(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
(vi)	(d) 120°
(vii)	(c) $(x + 2)$ or (d) $(x - 2)$ (any one correct)
(viii)	(c) above par
(ix)	(a) Both the statements are true.
(x)	(c) Both (A) and (R) are true, and (R) is correct reason for (A).
(xi)	(a) 15 cm
(xii)	(a) 3 : 4
(xiii)	(d) ₹1485
(xiv)	(a) {–5,–4,–3,–2,–1}
(xv)	(b) (A) is false, (R) is true.

Comments of Examiners

(i) Some candidates found the value of 'b^{2}' as 49 instead of 7 and chose the incorrect option (b). A few candidates used an incorrect formula for finding the discriminant.

(ii) Some candidates applied the formula for calculating simple interest and did not use the concept, Interest = MV – (P x n). A few candidates wrote option (c) too.

(iii) Many candidates chose option (d), 67%. They might not have been exposed to representing probability as percentages or visualizing 67% as a number greater than 1.

(iv) A few candidates did not have the concept that a line passing through the origin does not have x or y, hence chose option (a). Some candidates used the condition of intersecting lines, hence chose option (c).

(v) Some candidates squared each element individually instead of squaring the entire matrix, resulting in an incorrect answer option (d).

(vi) A few candidates wrote option (b) '60°' as the correct option due to the misconception that opposite angles of a cyclic quadrilateral are equal instead of supplementary. Some candidates did not use the condition that the base angles of an isosceles triangle are equal, hence chose option (a).

(vii) Most candidates got the answer to this question correct.

(viii) Many candidates assumed 10% as the market value and compared 12% and 10% and wrote option (b). A few candidates chose option (d) too.

(ix) Some candidates chose option (d) as they might only be familiar with the concept of circumcentre as the point of intersection of side bisectors. They might not have an understanding that it is also equidistant from the three vertices of a triangle. Several candidates got confused between incentre and circumcentre, thus chose option (b).

Suggestions for teachers

- Emphasize building mathematical reasoning, like identifying the nature of roots from the value of the discriminant.
- Encourage mathematical communication in the classroom. Ask students to explain different mathematical terms (Maturity value, interest, share above par, return percentage, GST, CGST, SGST, etc.) and their relationships by giving relevant examples.
- Use different representations (fractions, decimals, percentages) to show the probability of an event. Talk about why probability 'p' is in the range, $0 \leq p \leq 1$.
- Visualization of lines on a graph can help students enhance their understanding of lines.
- Demonstrate graphically that for parallel lines, the coefficient of x and the coefficient of y bear an equal ratio, lines passing through the origin are in the form $ax + by = 0$.
- Help students understand the unique nature of operations on matrices and not as procedural fluency or the application of standard algorithms.
- Focus on the misconception that squaring each element of a matrix will not yield the square of that matrix by giving relevant examples.
- Provide students with a variety of problems where they must apply the properties of circles and triangles to find the unknown angles.
- Demonstrate visually how the circumcentre is the point of intersection of the side bisectors and is equidistant from the three vertices of a triangle. Similarly, Incentre is the point of intersection of the angle bisectors of a triangle, equidistant from the three sides.
- Provide varied problems across all themes for applying one result to prove another result to strengthen assertion-reasoning questions.
- Help them deduce results like, 'For two similar objects, their area is proportional to the square of their sides and volume proportional to the cube of the sides' rather than mugging them up. Students must be fluent with cubes and cube roots of numbers up to 10.

(x) Many candidates were unable to relate the two statements (A) and (R) and wrote option (d).

(xi) Most candidates were able to answer this question. Some candidates incorrectly took the ratio AB : EF = AB : FG instead of AB : FE = BC : GF, resulting in the selection option (c).

(xii) Some candidates chose option (c) 9 : 16 due to confusion in the ratio of area and volume.

(xiii) Many candidates had taken the CGST percent as the GST percent, so they chose option (c). A few candidates did not read the question carefully. They calculated only the CGST and chose option (a).

(xiv) A confusion with the inequality signs made many candidates choose option (c) or option (d). Some candidates solved the inequality incorrectly and did not change the sign of the inequality when an inequality is multiplied or divided by a negative constant.

(xv) A few candidates divided 45 by 10 (might have included 0 in the set of Natural numbers) and chose option (c).

Suggestions for teachers

- Give practical exposure to students to find ratios of proportional parts of similar triangles by showing triangles in different orientations.
- Provide a variety of problems with different inequalities. Stress must be given on conditions like – $x < 6$ implies $x > -6$ by giving proper reasoning.
- Use number lines in representing sets for real numbers, integers, whole numbers and natural numbers and different solution sets.
- Give more practice problems based on assertions and reasons across different themes.

Question 2

(i) Solve the following quadratic equation $\mathbf{2x^2 - 5x - 4 = 0}$ **[4]**

Give your answer correct to **three significant figures**.

(Use mathematical tables for this question)

(ii) Mrs. Rao deposited **₹250** per month in a recurring deposit account for a period of **3** years. She received **₹10,110** at the time of maturity. Find: **[4]**

(a) the rate of interest.

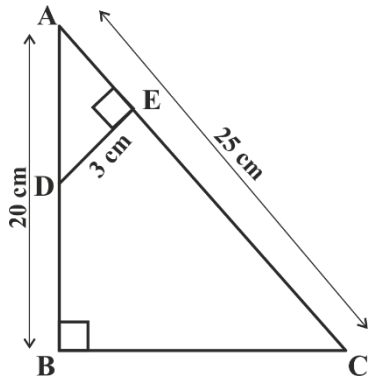
(b) how much **more** interest Mrs. Rao will receive if she had deposited **₹50** more per month at the same rate of interest and for the same time.

(iii) In **ΔABC, ∠ABC = 90°, AB = 20 cm, AC = 25 cm , DE** is perpendicular to **AC** such that **∠DEA = 90°** and **DE = 3 cm** as shown in the given figure. **[4]**

(a) Prove that **ΔABC ~ ΔAED.**

(b) Find the lengths of **BC**, **AD** and **AE**.

(c) If **BCED** represents a plot of land on a map whose actual area on ground is **576 m²**, then find the **scale factor** of the map.



MARKING SCHEME

(i)

$2x^2 - 5x - 4 = 0$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-4)}}{2(2)}$$

$$x = \frac{5 \pm \sqrt{25 + 32}}{4}$$

$x = \frac{5 \pm \sqrt{57}}{4} = \frac{5 \pm 7.550}{4}$ *or* $\frac{5 \pm 7.55}{4}$ ($\sqrt{57}$ =7.549 by finding square root)

$x = \frac{5 + 7.550}{4}$	$x = \frac{5 - 7.550}{4}$
$x = \frac{12.550}{4}$	$x = \frac{-2.550}{4}$
$x = 3.1375$	$x = -0.6375$
$x = 3.14$	$x = -0.638$

(ii)

P = ₹250, n = 36 months, MV = ₹10,110

(a) I = MV – (P × n) = 10,110 – 250 × 36

= 10,110 – 9,000

= ₹1,110

$$\frac{P.n(n+1)}{2} \times \frac{r}{100} \times \frac{1}{12} = 1110$$

$$\frac{\cancel{250^5} \times \cancel{36^3} \times 37 \times r}{2 \times \cancel{100_2} \times 12} = 1110$$

$$r = \frac{1110 \times 2 \times 2}{5 \times 3 \times 37} = 2 \times 2 \times 2$$

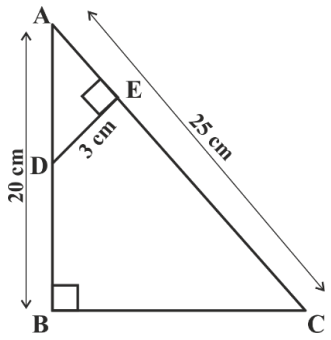
$r = 8\%$

(b) $\frac{50 \times 36 \times 37 \times 8}{2 \times 100 \times 12} = ₹37 \times 6$

= ₹222

(iii)

(a) In ΔABC and ΔAED



$\angle BAC = \angle EAD\ (common)$

$\angle ABC = \angle AED\ (each\ 90^o)$

Or, $\angle ADE = \angle ACB$

$\Delta ABC \sim \Delta AED$

$(b)\ \frac{AD}{AC} = \frac{AE}{AB} = \frac{DE}{BC}$

$BC = \sqrt{25^2 - 20^2} = \sqrt{225} = 15\ cm$

$\frac{AD}{25} = \frac{AE}{20} = \frac{\cancel{3}^{1}}{\cancel{15}_{5}}$

$AD = \frac{25}{5} = 5\ cm \quad and\ AE = \frac{20}{5} = 4\ cm$

(c) Area of quad BCED = $\frac{1}{2} \times 20 \times 15 - \frac{1}{2} \times 3 \times 4$

= 150 – 6 = 144 cm^2

$k^2 = \frac{area\ on\ map}{actual\ area}$

$= \frac{144\ cm^2}{576\ m^2}$

$= \frac{^{1}\cancel{144}\ cm^2}{_{4}\cancel{576} \times 100 \times 100\ cm^2}$

$\frac{1}{40000}$ ***or*** 1 : 40000

Scale factor = $\mathbf{\frac{1}{200}}$ ***or*** **1 : 200** ***or*** **200 : 1** ***or*** $\mathbf{\frac{200}{1}}$ ***or*** **200**

Comments of Examiners

(i) Some candidates used an incorrect formula to solve the quadratic equation, while a few made errors with the sign during substituting values in the formula.
Some candidates tried to find √57 by division method (instead of using Mathematical tables) and made calculation errors.
Many candidates made calculation errors while simplifying the expression. Some candidates made an error in rounding off the answer to 3 significant figures.

(ii) (a) Several candidates took the time as 3 years instead of 36 months. Many candidates calculated the rate of interest using the principal Rs 10,110. Some candidates have used an incorrect formula for the calculation of the rate of interest.

(b) Some candidates correctly found the interest (₹1332) using P = ₹300 but did not find the difference (₹1332 - ₹1110 = ₹222).

(iii) (a) The majority of candidates have correctly attempted this part. However, many candidates were unable to name the corresponding angles correctly to prove ΔABC ~ΔAED.

(b) The majority of the candidates found the value of BC correctly using the Pythagoras theorem. However, a few candidates wrote the proportional sides corresponding to equal angles incorrectly, for example, as

$$\frac{AD}{BC} = \frac{AE}{AC} = \frac{DE}{AB}$$

Hence, got the incorrect length of AD and AE.

(c) A very few candidates correctly found the scale factor k² (as the area was required). Many forgot to convert m² to cm², while some made errors in converting the area 576 m^2 to 576 ×100 ×100 cm^2.

Suggestions for teachers

- Work on building mathematical intuition in students. The instruction to 'use mathematical tables' indicates students should apply the quadratic formula. This hidden clue in the question is that the equation cannot be solved using regular factorization methods.
- Talk about how incorrect rounding off affects the result by taking relevant examples.
- Strengthen students' ability to understand the mathematical context and a clear grasp of mathematical language, like maturity value, rate of interest, for a period of three years, etc.
- Engage students with situational problems and apply mathematical reasoning effectively.
- Encourage students to logically build the formulae instead of mugging them up. Many students applying incorrect formulae show their lack of understanding of applied results/formulae.
- Give practical exposure in identifying and naming in proportional parts using similar triangles in different orientations.
- Focus should be given on naming the angles, for example ∠ AED cannot be written as angle ∠ E.
- Emphasise the concept of 'Scale factor' using various practical examples for better conceptual clarity. 'Scale factor' is a common term used in geography maps.
- Expose students to the interdisciplinary nature of concepts that cut across various disciplines.

Question 3

(i) Use ruler and compass for the following construction. Construct a ΔABC, where **AB = 6 cm**, **AC = 4.5 cm** and **∠BAC = 120°**. Construct a circle circumscribing the ΔABC. Measure and write down the length of the radius of the circle. **[4]**

(ii) If $\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{2} \\ \mathbf{3} & \mathbf{4} \end{bmatrix}$, $\boldsymbol{B} = \begin{bmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{4} & \mathbf{2} \end{bmatrix}$ and $\boldsymbol{C} = \begin{bmatrix} \mathbf{-5} & \mathbf{1} \\ \mathbf{7} & \mathbf{-4} \end{bmatrix}$ **[4]**

Find:

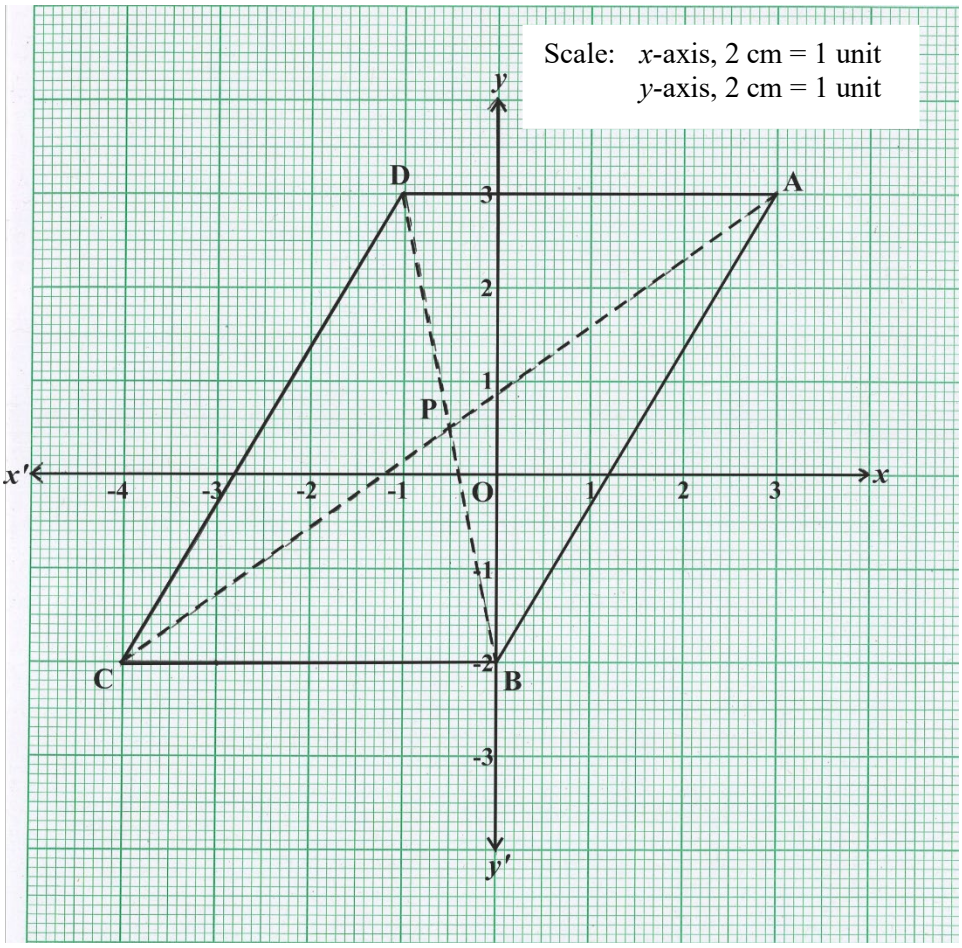
(a) A + C

(b) B(A+C)

(c) 5B

(d) B(A+C) – 5B

(iii) In the given graph **ABCD** is a parallelogram. **[5]**



Using the graph, answer the following:

(a) write down the coordinates of **A**, **B**, **C** and **D**.

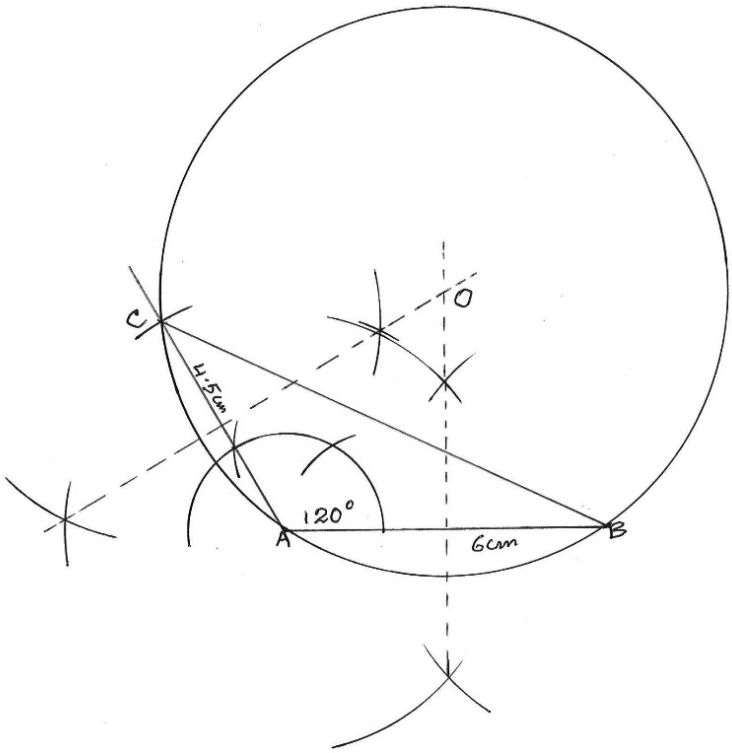
(b) **calculate** the coordinates of **'P'**, the point of intersection of the diagonals **AC** and **BD**.

(c) find the slope of sides **CB** and **DA** and verify that they represent parallel lines.

(d) find the equation of the diagonal **AC**.

MARKING SCHEME

(i)



AB or AC
$\angle A = 120^o$
ΔABC
Bisector of any one side
Circle with radius = 5.3 cm (± 0.2)

(ii)

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$

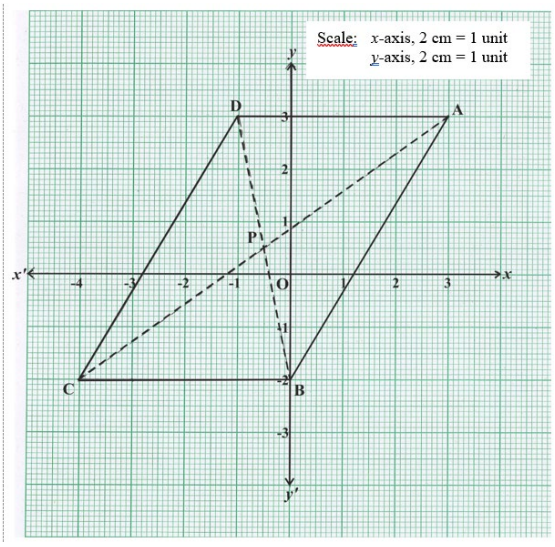
$(a)\ A + C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$

$= \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$

$(b)\ B(A + C) = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$

$= \begin{bmatrix} -8 + 10 & 6 + 0 \\ -16 + 20 & 12 + 0 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$

$(c)\ 5B = 5\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 20 & 10 \end{bmatrix}$

	$(d)\ B(A+C)-5B=\begin{bmatrix}2 & 6\\ 4 & 12\end{bmatrix}-\begin{bmatrix}10 & 5\\ 20 & 10\end{bmatrix}$ $=\begin{bmatrix}-8 & 1\\ -16 & 2\end{bmatrix}$
(iii)	$(a)\ A(3,3)\ \ B(0,-2)\ \ C(-4,-2)\ \ D(-1,3)$ $(b)\left(-\frac{1}{2},\ \ \frac{1}{2}\right)\ or\ (-0.5,0.5)\ or\ (-.5,.5)$ $(c)\ m_{CB}=\frac{-2-(-2)}{-4-0}=\frac{0}{-4}=0$ or, If AD ‖x-axis, slope = 0 $m_{DA}=\frac{3-3}{-1-3}=\frac{0}{-4}=0$ If BC ‖ x − axis, slope = 0 $\because m_{CB}=m_{DA}\ \ \therefore CB \parallel DA$ $\therefore AD \parallel BC$ $(d)\ m_{AC}=\frac{-2-3}{-4-3}=\frac{-5}{-7}=\frac{5}{7}$ $Eq\ of\ AC$ $y-3=\frac{5}{7}(x-3)$ $7y-21=5x-15$ $5x-7y-15+21=0$ $5x-7y+6=0$ ***equation in any equivalent form with integral co-eff.*** 

Comments of Examiners

(i) Some candidates used a protractor for constructing the angle 120° while several candidates bisected the angles and constructed the incircle instead of bisecting the sides to construct the circumcircle. A few candidates did the naming of the vertices of the triangle incorrectly.

(ii) Many candidates made errors when multiple arithmetic operations were involved in solving matrix-related problems. A few candidates found the matrix A + B instead of A + C. Some candidates made an error in matrix multiplication B (A + C). Some candidates did not multiply all the elements of matrix B by 5, while some made errors in matrix subtraction.

(iii) (a) Some candidates wrote as B (-2, 0) instead of B (0, -2).

(b) A few candidates wrote the *x*-coordinate of P and 0.5 instead of –0.5.

(c) Many candidates made mistakes in finding the slopes of lines AD and BC. Several of them wrote $0 \div 4 = 4$, and some left it as $\frac{0}{4}$. A few candidates found the slope correctly but did not verify that the lines were parallel.

(d) Candidates made a mistake in finding the slope as $-\frac{5}{7}$ instead of $\frac{5}{7}$. Hence, the equation of the line was incorrect.

Suggestions for teachers

- Encouraging students to begin with a rough sketch can help them visualize the problem better and plan the construction more effectively.
- Provide opportunities for peer-learning in constructing various angles using a ruler and a compass, the construction of triangles and naming them. Students can be asked to cross-check each other's work.
- Strengthen key skills in spatial reasoning and mathematical intuition. Help students understand the circumcircle and incircle through visual diagrams and provide opportunities to construct them.
- Highlight the correct order, along with the non-commutative nature of matrix operations. Build understanding that axioms and properties that apply to real numbers do not necessarily hold for matrices.
- The instructions should focus on helping students understand the unique nature of operations on matrices and not on procedural fluency or the application of standard algorithms.
- Encourage students to identify the coordinates of a point on the x and y axes as (x, 0) and (0, y) on a graph and how they are represented within the first brackets. Emphasise the sign while representing coordinates.
- Help students see slope not just as a formula, but as a meaningful concept tied to comparison, reasoning, and geometric interpretation.
- Providing opportunities to gain a deeper understanding of how such formulas are derived can help students better grasp the meaning of the variables involved in the general form of an equation.

SECTION B – (40 Marks)

*(Attempt **any four** questions from this **Section**.)*

Question 4

(i) Solve the following inequation, write the solution set and represent it on the real number line. **[3]**

$$2x - \frac{5}{3} < \frac{3x}{5} + 10 \leq \frac{4x}{5} + 11;\ x \in R$$

(ii) The first term of an Arithmetic Progression (A.P.) is **5**, the last term is **50** and their sum is **440**. Find: **[3]**

(a) the number of terms

(b) common difference

(iii) Prove that: **[4]**

$$\frac{(cotA + tanA - 1)(sinA + cosA)}{sin^3A + cos^3A} = secA.cosecA$$

MARKING SCHEME

(i) $2x - \frac{5}{3} < \frac{3x}{5} + 10 \leq \frac{4x}{5} + 11, \quad x \in R$

$2x - \frac{3x}{5} < 10 + \frac{5}{3}$

$\frac{10x - 3x}{5} < \frac{35}{3}$

$\frac{7x}{5} < \frac{35}{3}$

$x < \frac{35 \times 5}{7 \times 3}$

$x < \frac{25}{3}$

$x < 8\frac{1}{3}$

OR

$\frac{3x}{5} - \frac{4x}{5} \leq 11 - 10$

$\frac{3x - 4x}{5} \leq 1$

$-x \leq 5$

$x \geq -5$

$\frac{6x - 5}{3} < \frac{3x + 50}{5}$

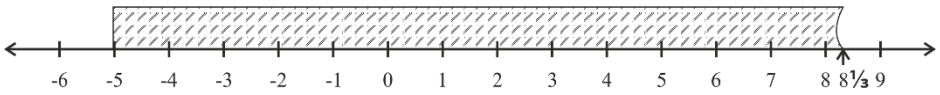
$30x - 25 < 9x + 150$

$30x - 9x < 150 + 25$

$21x < 175$

$x < \frac{175}{21}$

$x < 8\frac{1}{3}$

	Solution : $\{x: -5 \leq x < 8\frac{1}{3}, x \in R\}$ or $\frac{25}{3}$, *or* 8.3 
(ii)	(a) a = 5, T_n = 50, S_n =440 $T_n = a + (n-1)d$ $50 = 5 + (n-1)d$ $(n-1)d = 45$ $S_n = \frac{n}{2}\{2 \times 5 + (n-1)d\} = 440$ *or* $\frac{n}{2}(5+50) = 440$ $n\{10 + 45\} = 440 \times 2$ $n \times 55 = 440 \times 2$ $\therefore n = \frac{440 \times 2}{55} = \frac{40 \times 2}{5} = 8 \times 2 = 16$ (b) $(16-1)d = 45$ *or* $15d = 45$ $d = 3$
(iii)	$\frac{(cotA + tanA - 1)(sinA + cosA)}{(sin^3A + cos^3A)} = secA \cdot cosecA$ $L.H.S.$ $= \frac{\left(\frac{cosA}{sinA} + \frac{sinA}{cosA} - 1\right)(sinA + cosA)}{sin^3A + cos^3A}$ $= \frac{(cos^2A + sin^2A - sinAcosA)(sinA + cosA)}{(sin^3A + cos^3A)sinAcosA}$ $= \frac{sin^3A + cos^3A}{(sin^3A + cos^3A)sinAcosA}$ Or, $sin^2A + cos^2A = 1$ $= \frac{1}{sinAcosA} = secAcosecA = R.H.S.$

Comments of Examiners

(i) Some candidates made mistakes with signs while solving the two inequalities, especially when multiplying by a negative integer. Some candidates wrote the solution set in the roster form as {-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8}.

When representing the solution on the number line, many candidates overlooked one or more of the following:

- Arrows were not shown on the number line.
- Failed to mark at least one extra integer before –5 and after 8.3.
- failed to clearly mark and write 8.3 or $8\frac{1}{3}$ on the number line.
- Not shaded or drawn bold lines to show the solution set.

(ii) A few candidates got confused between the last term and the sum of the terms. The number of terms 'n' was taken as 50, and the 'n^{th}' term as 440; hence were unable to solve. A few candidates also used an incorrect formula for T_n and S_n.

Some candidates made calculation errors while applying the formulae 'n^{th} term = a + (n-1) d' and 'sum to n terms = $\frac{n}{2}$ [2a + (n-1) d]' instead of directly using the formula 'sum to n terms = $\frac{n}{2}$ (a + l)'

(iii) Many candidates failed to apply the algebraic identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ to factorize $Sin^3A + Cos^3A$. Most candidates tried to prove the identity by changing cot A = $\frac{cosA}{sinA}$, tan A = $\frac{sinA}{cosA}$ and by taking sin A. cos A as the LCM, but made errors.

Suggestions for teachers

- Emphasise on multiplying/dividing by a negative number in inequalities using various examples.
- Include visual representations like a number line often to show the solution sets for x∈ N, W, Z and R in the class. Discuss how real numbers are represented on a number line.
- Ask questions in the classroom that lead to discussions in the class, like how the solution set will change if the set N is changed to W, I or R, etc.
- Provide anti-examples along with examples. This would help students in developing a deeper understanding of the concept.
- Encourage students to read and understand questions and to carefully culling out essential information from the question.
- Help students to deduce formulae so that students are not mindlessly mugging them up and understand the meaning of them.
- Focus on developing problem-solving skills in students. Encourage students to apply multiple ways of approaching a problem. Encourage them to use different concepts that they previously learnt to reach new results.

Question 5

(i) Using properties of proportion, find the value of 'x': **[3]**

$$\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5} \; ; \; x \neq 0$$

(ii) It is given that $(x - 2)$ is a factor of polynomial $2x^3 - 7x^2 + kx - 2$. **[3]**

Find:

(a) the value of 'k'.

(b) **hence,** factorize the resulting polynomial completely.

(iii) A solid wooden capsule is shown in **Figure 1**. The capsule is formed of a cylindrical block and two hemispheres. **[4]**

Find the **sum** of total surface area of the three parts as shown in **Figure 2**. Given, the radius of the capsule is **3.5 cm** and the length of the cylindrical block is **14 cm**.

(Use $\pi = \frac{22}{7}$)

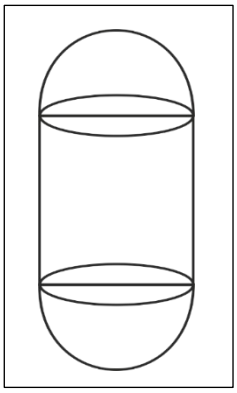


Figure 1

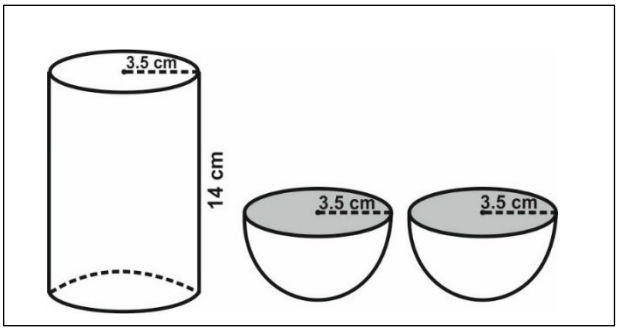


Figure 2

MARKING SCHEME

(i)

$$\frac{6x^2+3x-5}{3x-5}=\frac{9x^2+2x+5}{2x+5}$$

$$\frac{6x^2+3x-5+3x-5}{6x^2+3x-5-3x+5}=\frac{9x^2+2x+5+2x+5}{9x^2+2x+5-2x-5}$$

$$\frac{6x^2+6x-10}{6x^2}=\frac{9x^2+4x+10}{9x^2}$$

$$\frac{6x^2+6x-10}{2}=\frac{9x^2+4x+10}{3}$$

$$18x^2+18x-30=18x^2+8x+20$$

$$10x=50$$

$$\therefore x=5$$

(ii)

(a) $2x^3-7x^2+kx-2$ has $x-2$ a factor

$$\therefore 2\times 2^3-7\times 2^2+k\times 2-2=0$$

$$16-28+2k-2=0$$

$$\therefore 2k=14 \quad \therefore k=7$$

$$x-2\overline{)\,2x^3-7x^2+7x-2\,}\;\;2x^2-3x+1$$

$$-\,(2x^3-4x^2)$$

$$+\quad -3x^2+7x$$

$$+\,(-3x^2+6x)$$

$$-\quad x-2$$

$$-\,(x-2)$$

$$\times$$

(b) $2x^3-7x^2+7x-2$

$=(x-2)(2x^2-3x+1)$

$=(x-2)(x-1)(2x-1)$

(iii)

Total surface area =

Total surface area of cylinder + total surface area of 2 hemispheres

$$2\times\frac{22}{7}\times 3.5\times 14+2\times\frac{22}{7}\times 3.5^2+2\left\{\frac{4}{2}\times\frac{22}{7}\times 3.5^2+\frac{22}{7}\times 3.5^2\right\}$$

$\left(\pi=\frac{22}{7}\ or\ 3.14\right)$

$$=28\times\frac{22}{7}\times 3.5+2\times\frac{22}{7}\times 3.5^2+2\left\{3\times\frac{22}{7}\times 3.5^2\right\}$$

$$=22\times 14.0+8\times\frac{22}{7}\times 3.5\times 3.5$$

	$= 308 + 8 \times 22 \times 0.5 \times 3.5$ $= 308 + 308$ $= 616\ cm^2$

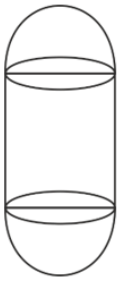


Figure 1

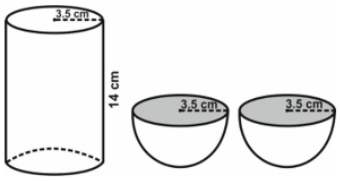


Figure 2

Comments of Examiners

(i) Many candidates solved the question through cross multiplication and not by applying the properties of proportion. Some candidates applied the concept of 'Componendo-Dividendo' but made errors with the sign.

(ii) (a) Most candidates were able to find the correct value of k by applying the Remainder and Factor Theorem, while a few candidates made calculation mistakes.

(b) Some candidates made errors in the long division or the factorization process. A few candidates did not answer in product form $(\boldsymbol{x-2})(\boldsymbol{x-1})(\boldsymbol{2x-1})$.

(iii) Many candidates failed to identify the formation of new surfaces on separating the cylinder and hemispheres.

Some candidates found the total surface area of the cylinder using 2πr (r + h), but did not find the total surface area of the two hemispheres. Some candidates found only the curved surface area of the hemisphere instead of the total surface area.

Candidates were confused with the formula of the total surface area of a cylinder and a hemisphere, and a few other candidates used incorrect formulae.

Suggestions for teachers

- Ask students to read and understand the question carefully.
- Encourage multiple ways of solving a question in the class. Talk about why a particular way of solving is more efficient than the other.
- Focus on the significance of using brackets while applying the concept of 'Componendo-Dividendo' to avoid sign errors.
- Focus on the interpretation of results to match what is asked in the question, like after factorizing a polynomial, the answer is to be written as the product of the factors.
- Hands-on practice should be given through 3-D models to identify different surfaces of a solid and a combination of solids.
- Facilitate students to construct different formulae instead of mugging them up. Nets, graph papers and other resources can be used to build conceptual understanding of solids and their formulae.

Question 6

(i) Use a graph paper for this question taking 2 cm = 1 unit along both axes. **[5]**

(a) Plot **A(1, 3)**, **B(1, 2)** and **C(3, 0)**.

(b) Reflect **A** and **B** on the ***x*-axis** and name their images as **E** and **D,** respectively. Write down their coordinates.

(c) Reflect **A** and **B** through the **origin** and name their images as **F** and **G** respectively.

(d) Reflect **A**, **B** and **C** on the ***y*-axis** and name their images as **J**, **I** and **H** respectively.

(e) Join all the points **A**, **B**, **C**, **D**, **E**, **F**, **G**, **H**, **I** and **J** in order and name the closed figure so formed.

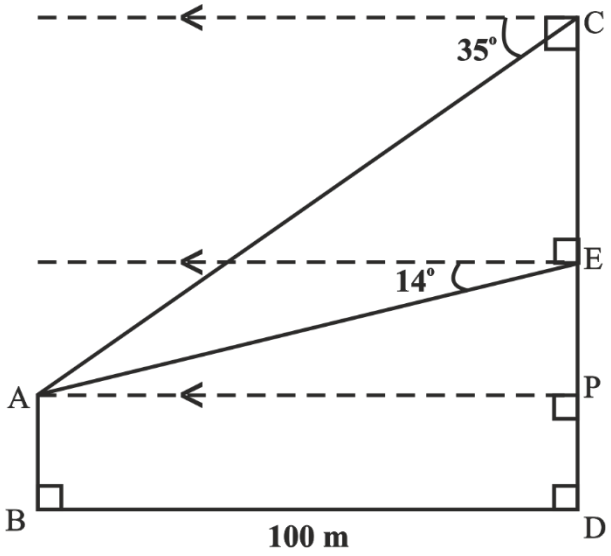
(ii) In the given diagram, **AB** is a vertical tower **100 m** away from the foot of a 30 storied building **CD**. The angles of depression from the point **C** and **E**, (E being the **mid-point** of CD), are **35°** and **14°** respectively. **[5]**

(Use mathematical table for the required values **rounded off** correct to **two places** of decimals only)

Find the height of the:

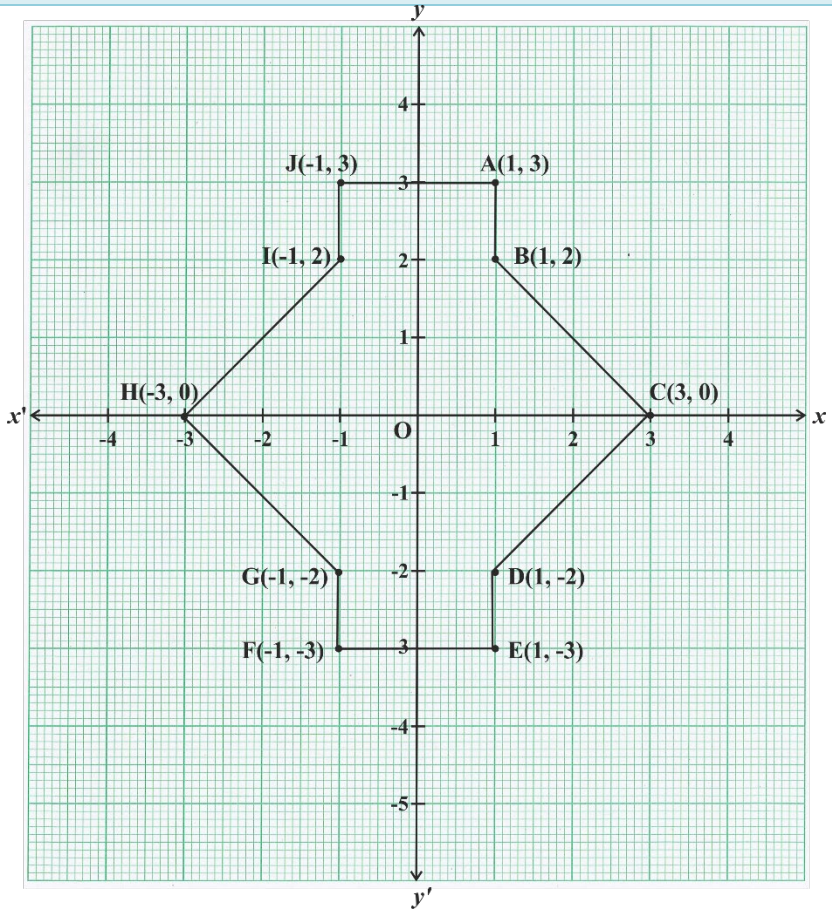
(a) tower **AB**

(b) building **CD**



MARKING SCHE

(i)



(a) Plotting A, B, C

(b) Reflecting A, B to E and D

(c) Reflecting A, B through the origin

(d) Reflecting A, B, C on *y*-axis **J, I and H**

(e) Decagon or 10-sided polygon

(ii)

$(a)\quad tan14° = \frac{EP}{100}$

$\therefore EP = 100 \times tan14°$

$= 100 \times 0.25$

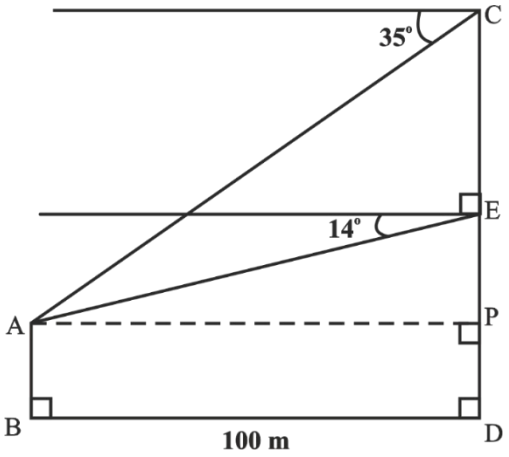
$= 25\ m$

$tan35° = \frac{CP}{AP} = \frac{CP}{100}$

$\therefore CP = 100 \times tan35°$

$= 100 \times 0.70$

$CP = 70\ m$



$\therefore CE = CP - EP = 70 - 25 = 45\ m$

$CE = ED = 45\ m$

$\therefore PD = DE - EP = 45 - 25 = 20\ m$ ⟶ A_1

$Height\ of\ tower\ (AB) = 20\ m$

Alternate Method:

Ht. $= \frac{DIST}{cot\alpha - cot\beta}$

$100 = \frac{CE}{Cot\ 55 - Cot\ 76}$

$\therefore$ CE = 100 (tan 35 – tan 14)

$= 100\ (0.70 - 0.25)$

$= 100\ \times\ 0.45$

$= 45\ m = ED$

$\therefore\ CD = 90\ m$

tan 14 $= \frac{EP}{100}$

$= 100 \times\ 0.25 = 25$ m

$\therefore$ AB = PD

$= 45 - 25$

$= 20\ m$

$(b)\ \ Height\ of\ building\ (CD) = 2CE = 2 \times 45 = 90\ m$

Comments of Examiners

(i) (a) Some candidates chose an incorrect scale, while some interchanged the positive and negative axes. A few candidates made mistakes in plotting and writing the coordinates. Several candidates plotted point C on the y-axis as (0,3) instead of plotting on x-axis.

(b) Most candidates plotted 'reflected A, B' points correctly on the x-axis, but some went wrong in writing the coordinates of the images or did not write the coordinates at all. Some reflected on the y-axis instead of on the x-axis.

(c) Most candidates reflected A and B correctly through the origin but did not name the points.

(d) A few candidates who went wrong in plotting the points A, B and C did incorrect plotting of images J, I and H.

(e) Very few candidates gave a correct name to the geometric figure formed. A few candidates did not join the points A, B, C, D, E, F, G, H, I and J.

(ii) (a) Some candidates used an incorrect ratio, for example, $\tan 14 = \frac{100}{EP}$ instead of $\frac{EP}{100}$. Candidates made mistakes in reading the value of tan 14° and tan 35° from the mathematical tables. A very common error among candidates was in rounding off to two decimal places. They rounded off from the initial working instead of at the end of the answer.

(b) Many candidates found an incorrect value of building CD, which is mainly due to an incorrect trigonometry ratio or incorrect rounding off.

Suggestions for teachers

- Provide more opportunities for reading, plotting, and naming the coordinates of points on the graph using various contexts, games, puzzles, and classroom activities.
- Use hands-on material like paper folding, a mirror, etc. to help students visualize the coordinates. Help students choose an appropriate scale to plot on the graph.
- Peer learning can be facilitated for students to develop a better understanding of the two axes and the four quadrants for plotting points.
- Provide opportunities for mathematical communication in class. Help students verbalize their mathematical ideas using appropriate mathematical terms. This would help them in naming geometrical figures also.
- Encourage students to draw and label the diagram themselves, marking the angle of depression from the horizontal to avoid confusing vertical and horizontal references.
- Talk about opposite and adjacent sides in various contexts.
- Focus on reading the trigonometric tables and rounding off the values.

Question 7

(i) Use a graph paper for this question. [3]

(Take 2 cm = 10 Marks along one axis and 2 cm = 10 students along another axis).

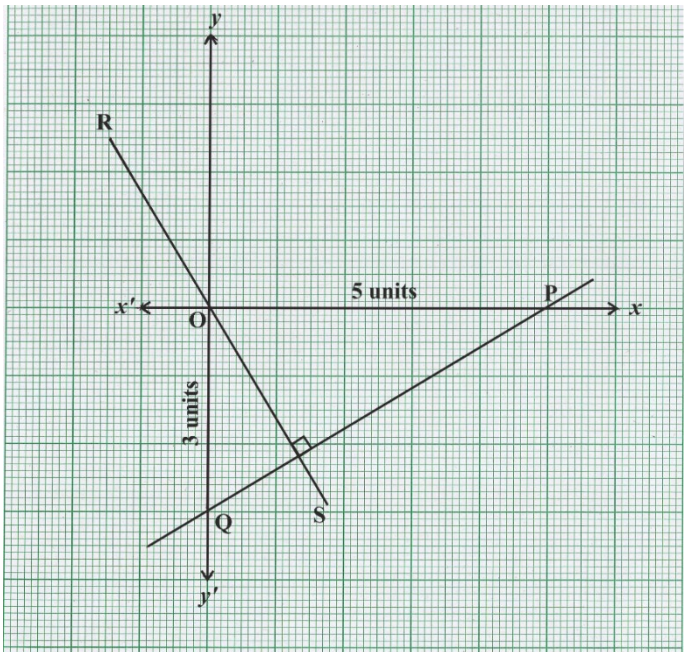
Draw a Histogram for the following distribution which gives the marks obtained by 164 students in a particular class and hence find the **Mode**.

Marks	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
Number of Students	10	26	40	54	34

(ii) In the given graph, **P** and **Q** are points such that **PQ** cuts off intercepts of **5 units** and **3 units** along the ***x*-axis** and ***y*-axis** respectively. Line **RS** is perpendicular to **PQ** and passes through the **origin**. Find the: [3]

(a) coordinates of **P** and **Q**

(b) equation of line **RS**



(iii) Refer to the given bill. [4]

A customer paid **₹2000** (rounded off to the nearest ₹10) to clear the bill.

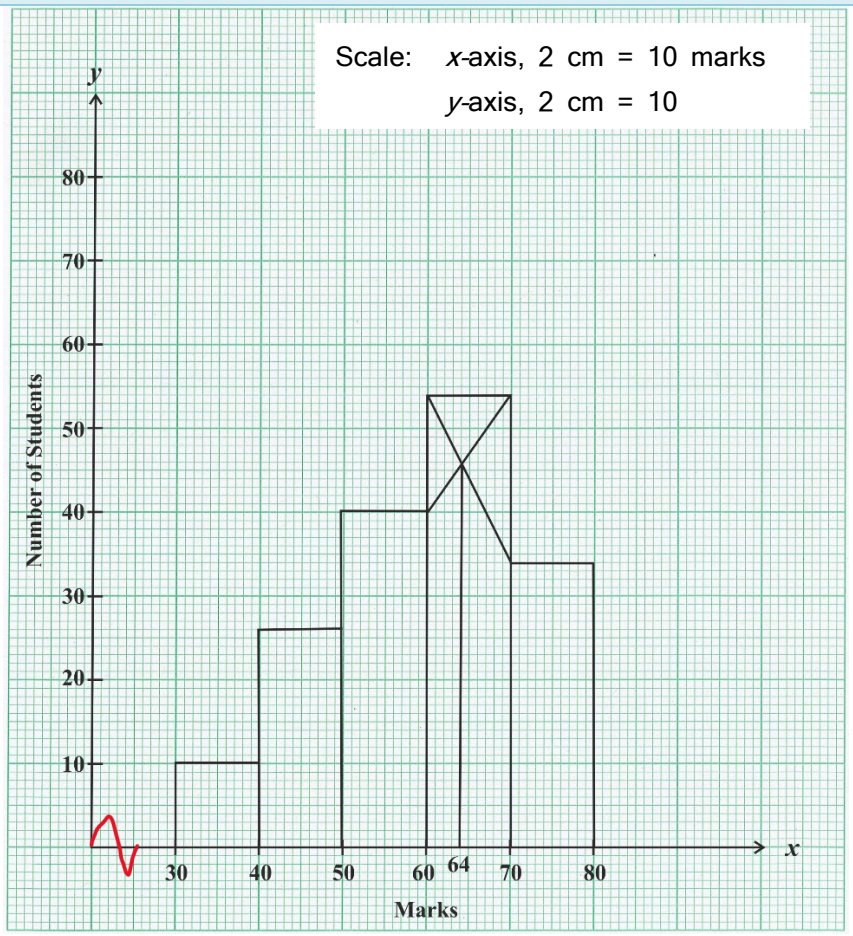
Note: 5% discount is applicable on an article if **10 or more** such articles are purchased.

BILL			
Article	**M.P. (₹)**	**Quantity**	**G.S.T.**
A	190	06	12%
B	50	12	18%

Check whether the total amount paid by the customer is correct or not. Justify your answer with necessary working.

MARKING SCHEME

(i)



Axis , kink & scale

Histogram

Mode 64 ± 1

(ii) (a) P(5, 0), Q(0, –3)

$(b)\ Slope\ of\ PQ: \frac{-3-0}{0-5} = \frac{-3}{-5} = \frac{3}{5}$

$\therefore Slope\ of\ RS: -\frac{5}{3}$

Equation of RS passing through (0, 0) is.

$y - 0 = -\frac{5}{3}(x - 0)$

$3y = -5x \quad or \quad 5x + 3y = 0$

(iii) A: MP = ₹190 No. of articles = 6 Total price = 190 × 6 = ₹1140

	$GST = 1140 \times \frac{12}{100} = \frac{13680}{100} = ₹\mathbf{136.80}$ Total price of A $= 1140 + 136.80 =$ ₹1276.80 B: MP = 50 No. of articles = 12, Discount = 5% $\therefore Total\ price = 50 \times 12 - 50 \times 12 \times \frac{5}{100}$ $= 600 - 30 =$ ₹570 GST $= 570 \times \frac{18}{100} = ₹\mathbf{102.60}$ Total price of B $= 570 + 102.60 =$ ₹672.60 Bill amount $= 1276.80 + 672.60 =$ ₹1949.40 **Rounded off to nearest ₹10 = ₹1950.00** The total amount paid by the customer is **incorrect.**

Comments of Examiners

(i) Common errors made by candidates in drawing a histogram and finding the mode included:

- Use of an incorrect scale
- Failure to show a kink when the graph's first point is at 30.
- Labelling of the x-axis as 30 – 40, 40 – 50…., instead of each end point of class boundaries as 30, 40 …
- Incorrect plotting of the graph.
- The gaps between the class intervals were not uniformly marked.
- Incorrect identification line for the mode.
- Having a correct graph, but unable to read the mode value accurately.

(ii) (a) Some candidates did not write the coordinates of P and Q correctly, for example, writing as P (0, 5), Q (-3, 0) instead of P (5, 0), Q (0, –3).

(b) Candidates made an error in finding the slope of the line RS perpendicular to PQ. Some candidates did not take (0,0) to find the equation of RS, but instead took points that do not lie on the line. Many candidates gave the final answer of the equation with a fractional coefficient as $y=-\frac{5}{3}x$ instead of 3y = -5x or 5x + 3y = 0.

(iii) Many candidates missed out the given condition of the discount and calculated the bill without using it. Several candidates did not round off the final answer.

Suggestions for teachers

- Provide hands-on practice with graph paper, stressing equal spacing of intervals and precise plotting so that students understand the importance of uniformity.
- Provide several problems where students select an appropriate scale themselves to represent the data efficiently.
- Explain the purpose of showing a 'kink' when data does not start from zero, rather than asking them to memorize the symbol.
- Encourage students to cross-check their answers by estimation (mode should be near the highest bar) to develop number sense and avoid mechanical misreading of the graph.
- Use graph paper or geo-board activities where the students plot various points.
- Provide visual demonstrations on graph paper to make the properties of lines, points, and slopes intuitive rather than formula driven. Applications like GeoGebra should be used to visualize the manipulate the data.
- Include actual shopping bills or project works in the classroom so students see how discounts, GST, and rounding off work in everyday transactions.

Question 8

(i) A man bought **₹200** shares of a company at **25%** premium. If he received a return of **5%** on his investment. Find the: [3]

(a) market value

(b) dividend percent declared

(c) number of shares purchased, if annual dividend is **₹1000.**

(ii) For the given frequency distribution, find the: [3]

(a) mean, to the nearest whole number

(b) median

x	10	11	12	13	14	15	16
f	3	2	2	6	3	5	3

(iii) Mr. and Mrs. Das were travelling by car from Delhi to Kasauli for a holiday. Distance between Delhi and Kasauli is approximately **350 km** (via NH 152D). Due to heavy rain they had to slow down. The average speed of the car was reduced by **20 km/hr** and time of the journey increased by **2 hours**. Find: [4]

(a) the original speed of the car.

(b) with the reduced speed, the number of hours they took to reach their destination.

MARKING SCHEME

(i)

$(a)\ M.V.\ of\ shares = 200 + \frac{25 \times 200}{100} = ₹250$

$(b)\ Dividend = 200 \times \frac{r}{100} = 2r$

$Return\ \% = 5 \qquad 250 \times \frac{5}{100} = 2r \quad \therefore r = \frac{25}{4} or\ 6\frac{1}{4}\%$ or 6.25

$(c)\ Number\ of\ shares = \frac{1000}{25/2} = 80\ shares$

(ii)

x	f	fx	Cf
10	3	30	3
11	2	22	5
12	2	24	7
13	6	78	13
14	3	42	16
15	5	75	21
16	3	48	24
	$\sum f$=24	$\sum fx$=319	

$Mean = \frac{319}{24} \qquad Mean = 13.2 = 13\ (nearest\ whole\ number)$

$Median = 13$

(iii)

$\frac{350}{x}$ **OR** $\frac{350}{x-20}$

$\therefore \frac{350}{x-20} - \frac{350}{x} = 2$

$350x - 350x + 7000 = 2x^2 - 40x$

$2x^2 - 40x - 7000 = 0$

$x^2 - 20x - 3500 = 0$

$(x - 70)(x + 50) = 0$

$\therefore x = 70$

$(a)\ Original\ Speed\ = 70\ km/hr$

$(b)\ Time\ with\ reduced\ speed = \frac{350}{70-20} = \frac{350}{50} = 7\ hrs$

Comments of Examiners

(i) (a) Some candidates did not understand the meaning of shares at a 25% premium and got the incorrect Market value of shares.

(b) Many candidates did not understand the term 'return of 5%' in the given context. Some candidates used market value instead of face value to calculate the rate.

(c) Many candidates calculated the number of shares incorrectly. They lack understanding of the terminology and context used in the question.

(ii) (a) Some candidates made a calculation error in finding fx. A few candidates did not round off the answer to the nearest whole number. A few candidates converted it to a grouped frequency distribution, which did not tally with the given data, and hence got an incorrect answer to the mean.

(b) Many candidates were unable to answer this question. Some incorrectly took the average of the frequencies to find the median. A few candidates found the median using the average of the median ranks, while others determined the median without showing the cumulative frequency (Cf).

(iii)(a) Some candidates were unable to identify the bigger fraction between $\frac{350}{x}$ and $\frac{350}{x-20}$, hence the equation formed was incorrect. Some went wrong in factorizing the quadratic equation $x^2 - 20x - 3500 = 0$

(b)A few candidates found only the reduced speed and did not calculate the number of hours.

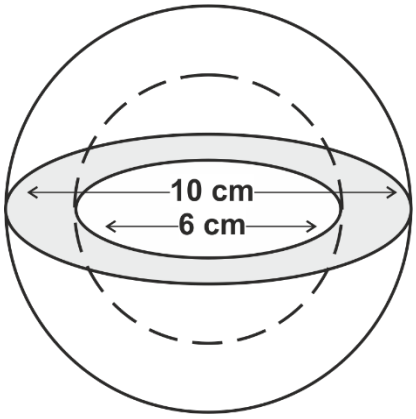
Suggestions for teachers

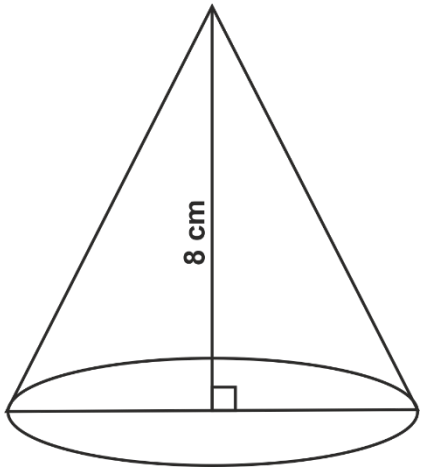
- Focus on mathematical communication through real-life examples (premium/discount, interest vs return) and use visual comparisons (tables, charts) rather than repeated practice.
- Encourage reasoning, not memorization. Small stepwise problems and reflective questioning can help students internalize the concept, like why the dividend is declared on face value, but the return is calculated on the actual investment, etc.
- Encourage students to express the improper fractional answers in mixed fractions.
- Give a variety of problems for finding the median for all types of data (raw, discrete, grouped, and continuous grouped) to the students.
- Encourage reasoning in classrooms, like lower speed means more time, so $\frac{350}{x-20} = \frac{350}{x} + 2$. Use simple, relatable examples (e.g., walking faster/slower) to highlight this relationship.
- More practice for factorization should be provided to the students.

Question 9

(i) A hollow sphere of external diameter **10 cm** and internal diameter **6 cm** is melted and made into a solid right circular cone of height **8 cm**. Find the radius of the cone so formed. [3]

[Use $\pi = \frac{22}{7}$]





(ii) Ms. Sushmita went to a fair and participated in a game. The game consisted of a box having number cards with numbers from **01 to 30.** The three prizes were as per the given table: [3]

Prize	**Number on the card drawn at random is a**
Wall Clock	perfect square
Water Bottle	even number which is also a multiple of 3
Purse	prime number

Find the probability of winning a:

(a) Wall Clock

(b) Water Bottle

(c) Purse

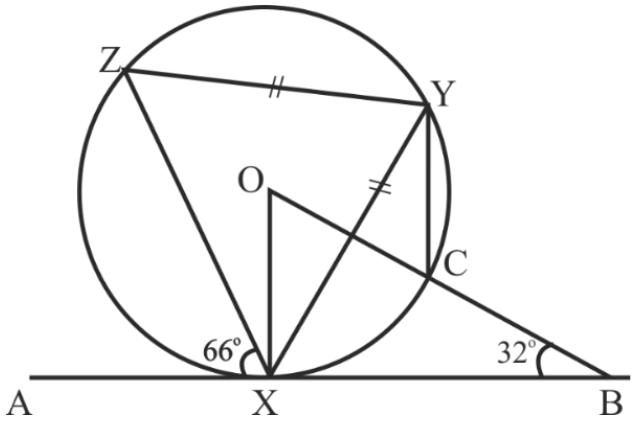
(iii) **X**, **Y**, **Z** and **C** are the points on the circumference of a circle with centre **'O'**. **AB** is a tangent to the circle at **'X'** and **ZY = XY**. Given ∠**OBX = 32°** and ∠**AXZ = 66°**. Find: [4]

(a) ∠BOX

(b) ∠CYX

(c) ∠ZYX

(d) ∠OXY



MARKING SCHEME

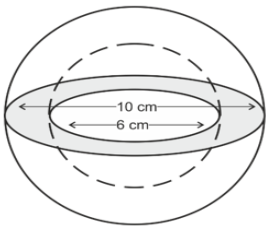
(i) $Volume\ of\ hollow\ sphere\ =\ Volume\ of\ Cone$

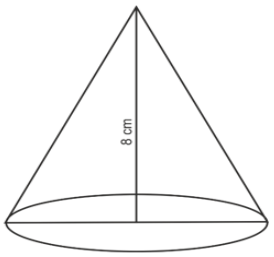
$\frac{4}{3}\times\frac{22}{7}(5^3-3^3)=\frac{1}{3}\times\frac{22}{7}(r^2)\times 8.$ $\left(\pi=\frac{22}{7}\ or\ 3.14\right)$

$4\times(125-27)=8r^2$

$\therefore r^2=\frac{4\times 98}{8}=49$

$\therefore\ r=\ 7cm$





(ii) $(a)\{1,4,9,16,25\}$

$Probability\ (Wall\ Clock)=\frac{5}{30}=\frac{1}{6}$

$(b)\ \{6,12,18,24,30\}$

$Probability\ (Water\ Bottle)=\frac{5}{30}=\frac{1}{6}$

$(c)\ \{2,3,5,7,11,13,17,19,23,29\}$

$Probability\ (Purse)=\frac{10}{30}=\frac{1}{3}$

(iii) (a) $\angle BOX = 180° - (90° + 32°) = 58^o$ (sum of angles of a Δ = 180º Angle made by tangent and radius = 90º)

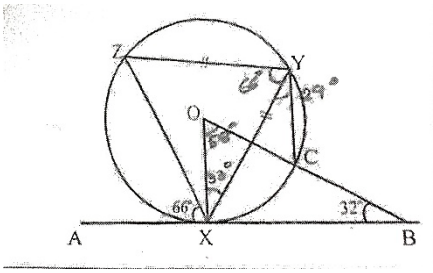
(b) $\angle CYX = \frac{1}{2}\angle COX = \frac{1}{2} \times 58° = 29°$ (angle at the centre is double angle in the remaining circumference)

(c) $\angle ZYX = \angle AXZ = 66°$ (alternate segment theorem)

(d) $\angle YZX = \angle YXZ = \frac{180° - 66°}{2} = \frac{114°}{2} = 57°$ (Isosceles Δ)

$\therefore \angle BXY = \angle YZX = 57°$

$\therefore \angle OXY = 90° - 57° = 33°$



Comments of Examiners

(i) A few candidates used an incorrect formula for finding the volume of a hollow sphere or a solid cone. Some candidates took the given diameters as the radii. Simple calculation errors were also common.

(ii) (a) Some candidates did not include 1 as a perfect square, hence made a mistake in finding the probability.

(b) The majority of candidates answered this question correctly.

(c) Many candidates considered 1 as a prime number, which led to an incorrect answer.
A few candidates did not simplify the fractions obtained.

(iii) (a) A few candidates did not mention the reasoning.

(b) Some candidates incorrectly wrote ∠BOX = ∠CYX, assuming them to be angles in the same segment.

(c) The majority of candidates answered this question correctly.

(d) Many candidates did not use the properties of isosceles triangles and the theorem on the angle between a tangent and a radius, leading to incorrect answers.

Suggestions for teachers

- Using real-life analogies (e.g., melting metal to recast) builds understanding that the volume remains the same when one object is molded into another.
- Engage students in deriving the formula for a hollow sphere by subtraction of two solid spheres, so they conceptually see why both radii are needed.
- Help students list out all possible favorable outcomes before finding the probability. This will reduce errors in identifying outcomes.
- Remind students to express the probability in its lowest fractional form.
- Engage students in meaningful discussions in the class, like why 1 is excluded from the set of primes but included in perfect squares, etc.
- Emphasise precise angle representation and labelling and providing reasons for the inferences drawn.
- Encourage peer learning and visual reasoning (dynamic geometry software like GeoGebra, paper folding or physical models) to observe various results and theorems on geometry. This will improve comprehension and reduce memorization of results.

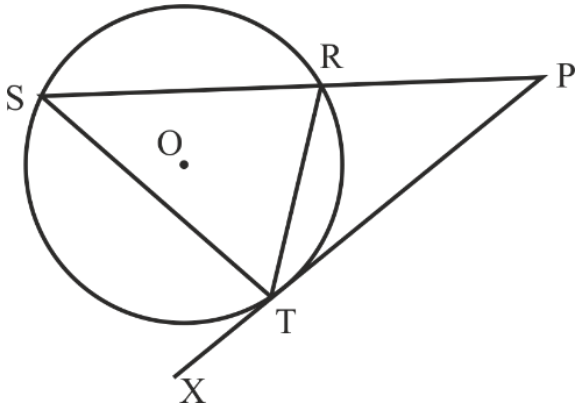
Question 10

(i) If **1701** is the nth term of the Geometric Progression (G.P.) 7, 21, 63 ……., find: [3]

(a) the value of ***'n'***

(b) hence find the **sum** of the ***'n'*** terms of the G.P.

(ii) In the given diagram **'O'** is the centre of the circle. Chord **SR** produced meets the tangent **XTP** at **P**. [3]

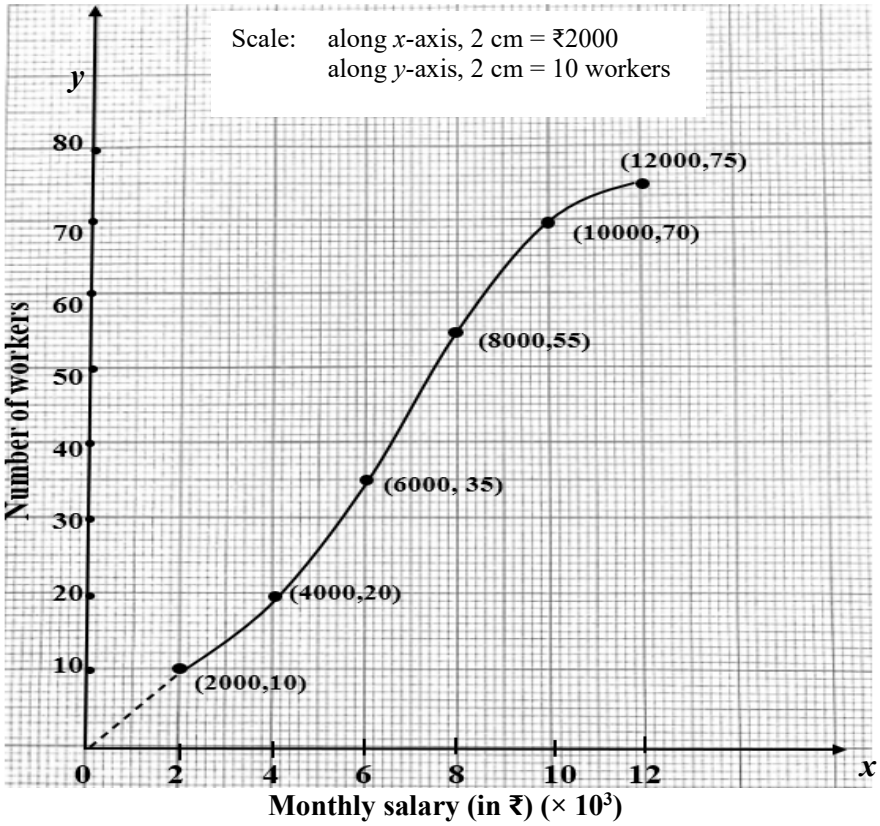


(a) Prove that **ΔPTR ~ ΔPST**

(b) Prove that $\mathbf{PT^2 = PR \times PS}$

(c) If **PR = 4 cm** and **PS = 16 cm**, find the length of the tangent **PT**.

(iii) The given graph represents the monthly salaries (in ₹) of workers of a factory. [4]



Using graph answer the following:

(a) the total number of workers.

(b) the median class.

(c) the lower-quartile class.

(d) number of workers having monthly salary more than or equal to ₹6,000 but less than ₹10,000.

MARKING SCHEME

(i)

$7, 21, 63, \ldots\ldots\ldots\ldots$

$(a)\ T_n = 1701$

$a = 7, \quad r = 3 \quad T_n = ar^{n-1}$

$\therefore 1701 = 7 \times 3^{n-1}$

$243 = 3^{n-1}$

$3^5 = 3^{n-1} \quad \therefore n - 1 = 5 \quad so, n = 6$

$(b)\ S_6 = \dfrac{a(r^n - 1)}{r - 1}$

$= \dfrac{7(3^6 - 1)}{2} = \dfrac{7 \times (729 - 1)}{2} = \dfrac{7 \times 728}{2}$

$= 2548$

(ii)

(a) *In* ΔPTR *and* ΔPST

$\angle P$ *is common to both*

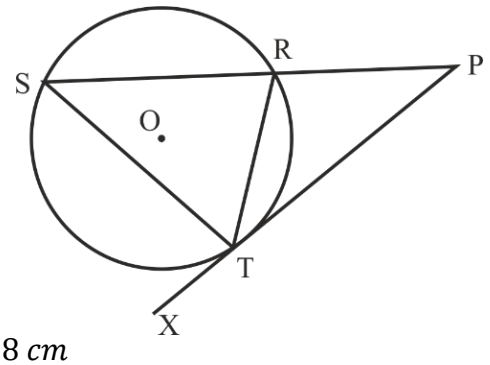
$\angle PTR = \angle PST$ *(alt segment)*

$\therefore\ \Delta PTR \sim \Delta PST$

$(b)\ \therefore \dfrac{PT}{PR} = \dfrac{PS}{PT}$ *Hence* $PT^2 = PR \times PS$

$(c)\ PT^2 = 4 \times 16 \quad or \quad PT = 2 \times 4$

$PT = 8\ cm$



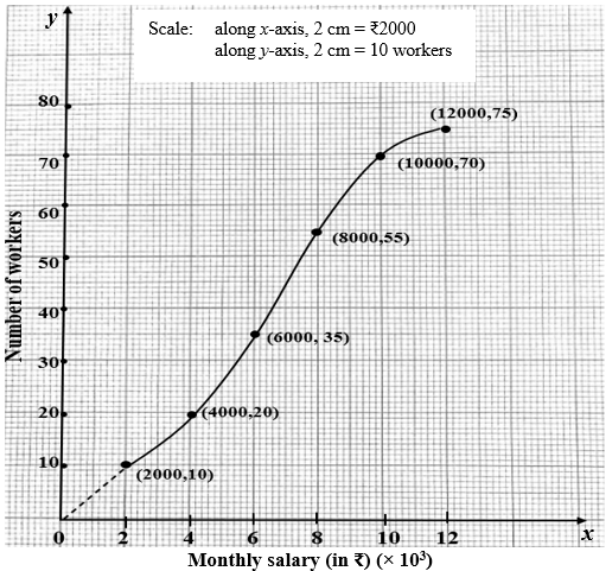
(iii)

(a) Total number of workers = 75

(b) ₹6000 – ₹8000

(c) ₹2000 – ₹4000

(d) 70 – 35 = 35


Scale: along x-axis, 2 cm = ₹2000
along y-axis, 2 cm = 10 workers
y
x
Number of workers
Monthly salary (in ₹) (× 10³)
(2000,10)
(4000,20)
(6000, 35)
(8000,55)
(10000,70)
(12000,75)
0
2
4
6
8
10
12
10
20
30
40
50
60
70
80

Comments of Examiners

(i) (a) Some candidates were confused with the formula of A.P. and G.P. A few used incorrect formulae of the nth term of G.P. Many candidates wrote 1701= 7× 3n-1 = 21n-1.
Few candidates found r = $\frac{7}{21} = \frac{1}{3}$ instead of r = 3.

(b) Few candidates wrote 3^6=243 instead of 729. Some used an incorrect formula for the sum of G.P, while some made mistakes in finding the sum due to an error in finding the nth term.

(ii) (a) Candidates were unable to prove ΔPTR ~ ΔPST. They were unable to identify two equal angles. Some candidates wrote ∠T = ∠S instead of ∠PTR = ∠PST.

(b) Many candidates made mistakes in writing the corresponding proportional sides.
Some candidates directly stated the theorem $PT^2 = PS \times PR$ without proving that the corresponding sides of similar triangles are in proportion.

(c) Candidates went wrong in finding the value of PT due to errors in part (b).

(iii) (a) Some candidates were not very familiar with reading values from the graph and wrote incorrect values for the number of workers. Many candidates wrote the answer as 80 instead of 75.

(b) Some wrote the answer as ₹6 - ₹8 instead of ₹6000 - ₹8000. Some candidates tried finding the median value instead of the median class.

(c) Some wrote the answer as ₹2 - ₹4 instead of ₹2000 - ₹4000. Some candidates mentioned the lower quartile instead of the lower quartile class.

(d) A few candidates made calculation mistakes.

Suggestions for teachers

- Help students differentiate A.P. and G.P. through the idea of addition vs. multiplication using patterns.
- Reinforce students to derive and understand a formula instead of memorizing it.
- Use small exponents to build intuition (3^5 = 243, 3^6=729) before applying to bigger values.
- Encourage visual reasoning (dynamic geometry software like GeoGebra, paper folding or physical models) to observe various results and theorems on geometry. This will improve comprehension and reduce memorization.
- Focus on the naming of angles using three points by explaining why it is helpful.
- Use classroom activities where students construct the graphs from a frequency table. Help students to carefully read cumulative frequency graphs, emphasizing scale interpretation (e.g., ₹2000 not "2" on x-axis, 10 workers not "2" on y-axis).
- Provide examples to stress the difference between median value vs. median class, and lower quartile value vs. lower quartile class.
- Geometry demands strong spatial reasoning and conceptual understanding. Use hands-on activities, visual tools, sketches and real-life applications to make geometrical ideas more engaging.

GENERAL COMMENTS

Topics found difficult by candidates.

- SGST, CGST, GST
- Maturity value & interest formula
- Premium price of shares
- Plotting accurately on number line as per the replacement set
- Solving quadratic equation
- Matrix multiplication
- Dividing a polynomial by a binomial
- T_n and S_n of A.P. & G.P.
- Naming the geometric figure
- Finding the equation of a line
- Finding other components after proving triangles similar.
- Locus constructions, construction of circumcircle.
- Using properties of circles
- Heights and distances
- Sum involving non-standard angles.
- Mensuration formulae for split shapes
- Reading value from the given ogive
- Fewer candidates attempted Q9 and Q10, which included geometry questions as sub-parts, indicating a tendency to avoid geometry-based problems. Many candidates seem to prefer numerical questions that involve direct computation, which they perceive as easier and familiar.

Concepts in which candidates got confused.

- SGST, CGST, GST
- Maturity Value & Interest Formula
- Preimum price and market price of shares
- Set of real numbers and set of integers/whole Numbers/Natural Numbers.
- T_n and S_n of A.P. & G.P.
- Mensuration formulae for curved surface area, total Surface Area, Volume
- Concept of circum-centre and in-centre
- circles and naming angles
- Corresponding parts of similar triangles.

Suggestions for candidates

- Read all the instructions and questions carefully, with special attention to multiple-choice questions. Evaluate all options before answering. Avoid showing the working of multiple-choice questions.
- Utilize the reading time in reading and selecting the questions from the given options.
- Each subpart must be answered. The solution should be concise and to the point. Be methodical and neat in working.
- All steps of working, along with the rough work, must be shown on the same page where the question is answered.
- Provide proper reasons for geometrical property-based answers, like relevant statements of theorems and properties. Do not skip important steps.
- In Geometric constructions, all traces and arcs must be clearly shown.
- Round off answers where required, ensuring that all rounding is performed only at the final stage. Also, try to give the final answer in the simplest fraction.
- Use Mathematical tables to find square roots, values of non-standard trigonometric angles.
- Give extra attention to the change of sign while working with inequalities.
- Use graph paper for graph-related questions and follow the scale provided in the question.
- Solve previous years' papers as well as the specimen paper provided by the ICSE board on the CISCE website [www.cisce.org]
- Make note of all the important formulae, identities, results, and theorems for a quick revision.
- Use the last 10-15 minutes to review the answers.

Suggestions for Teachers

- Encourage **mathematical thinking** capacities in students, such as mathematical reasoning, working with abstraction, observing patterns, and identifying important cues in problems. Build intuition by helping students understand underlying concepts rather than relying on rote procedures.
- Promote mathematical **visualization skills**, especially for abstract ideas. Students often avoided geometry questions, indicating a need for more opportunities to visualize spatial relationships through sketches, hands-on activities, or real-life applications to strengthen understanding.
- Focus on **conceptual understanding** over procedures. Instead of merely following an algorithm or procedure, greater emphasis should be placed on interpreting and making sense of the underlying abstract ideas.
- Develop students' ability to understand and use precise mathematical vocabulary for **mathematical communication**.
- Strengthen **comprehension of variables** and their roles in formulae/equations/general representations. Emphasis on understanding the process of arriving at a generalization or deriving a formula during the teaching–learning process has a greater impact on students' grasp of abstract ideas in mathematics than merely presenting ready-made formulae.
- Use **interdisciplinary approaches** to make abstract concepts more relatable and help students connect mathematical ideas across disciplines, enhancing engagement and understanding.